



CARE Radiology Reporting Workspace

Architecture Audit Report

Deep code audit of the complete radiology reporting workflow

Repository: DrAbinash/care-on-synology1

HEAD: d4515dae (PR #658 branch)

Overall Maturity Score

7.2 / 10

NEEDS TARGETED HARDENING

Ready for clinic test with caveats

1. Executive Summary Dashboard

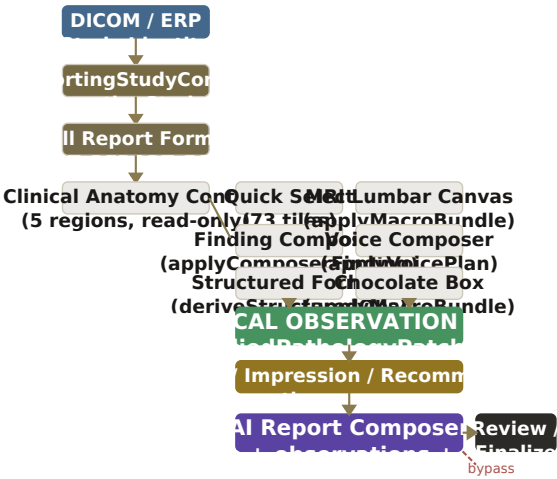
Maturity Assessment

Dimension	Score	Status	Key Finding
Observation Ledger Convergence	8/10	CONNECTED	5 producers write to ledger; AI Composer apply bypasses it
AI Composer Context	9/10	CONNECTED	Canonical study context + observations + persona all wired
Freshness / Safety	9/10	SAFE	inputHash + reportRevision dual-axis; STALE_READY works
Slot Identity	8/10	CONNECTED	region concept level laterality; voice conflictGroup divergence risk
Manual Text Protection	9/10	SAFE	Provenance-based; protected flag; undoLastPatch
Save / Reopen Persistence	8/10	CONNECTED	serializeObservationLedger + hydrate with reconciliation
Structured Reporting → Ledger	7/10	PARTIAL	Adapter wired; toggle-off does not remove observations
Clinical Anatomy Context	7/10	CONNECTED	5 regions; read-only; not cross-validated vs observationSlot
MRI Format Crosswalk	8/10	CONNECTED	7 EXACT + 3 CLOSE + 7 ADDED; 320 clinic files audited
Finalize / Output Safety	9/10	SAFE	Finalized blocks all mutations; STALE_READY blocks blind apply
Test Coverage	8/10	CONNECTED	214 er-side + 5 api-server test files; 2401 tests pass

Strengths, Gaps, Duplications & Technical Debt

STRENGTHS	GAPS	DUPLICATION	TECH DEBT
Single canonical ledger (appliedPathologyPatches) for all observation producers	AI Composer apply bypasses ledger — no post-apply reconciliation	voiceComposerObservations[] mirrors voice-* patches in ledger	RadiologyReportingWorkspa ce.tsx is 5712 lines — monolith
Canonical slotKey (region co ncept level laterality) with same-slot replacement	Persona router does NOT exist — single generic system prompt	appliedFormatName vs appliedFormatReportTitle — two display fields	Quick Select tile ids are non-deterministic (uid() at module load)
PR #654 adapter: deriveComposeObservations feeds AI from ledger	Structured toggle-off does NOT remove ledger observation	3 independent region→bodyPart maps (reportingStudyContext, clinicalAnatomy, pickStructuredTemplate)	ABNORMALITY_HINTS list is hardcoded (19 terms, no config)
PR #656 freshness: inputHash + reportRevision dual-axis	root.exiting/root.traversing collapse into foraminal_stenosis		CONCEPT_CANON not cross-validated vs clinicalAnatomyContext
Clinical Anatomy Context: 5 regions, clinic-format-driven ordering	Cervical/Dorsal spine just shallow-clone LS Spine (shared refs)		

2. End-to-End Clinical Workflow



Red dashed line: AI Composer accepted draft bypasses the canonical ledger (writes narrative only).

3. Data & State Ownership

Finding Producer Convergence Table

Producer	Function	Writes to Ledger?	Path
Quick Select (findings)	applyPathologyOverlay	YES	store.ts:981
Quick Select (history/tech/recommendations)	mergeField	NO	store.ts:setField
Finding Composer	applyComposerFinding	YES	store.ts:1437
Structured Format	deriveStructuredObservations + applyMacroBundle	YES	structuredObservations.ts
Chocolate Box (bundle)	applyMacroBundle	YES	store.ts:1182
MRI Lumbar Canvas	applyMacroBundle	YES	workspace.tsx:4516
Voice Composer	applyVoiceComposerPlan	YES (voice-* patches)	store.ts:1661
AI Composer (apply)	applyAiComposerAccepted	NO — narrative only	store.ts:1719
Full Report Format	confirmOverwriteAndApply	Marks stale, no new patches	store.ts:897
Snippet Macros	applyMacroWithValues	NO — field only	workspace.tsx
Manual typing	setField(source:'manual')	NO — field only	store.ts

4. Observation Ledger Flow



Slot Key & Same-Slot Replacement Semantics

The canonical slot identity is region | concept | level | laterality. Severity, measurement, source, conflictGroup, and anatomicalSection are intentionally NOT part of the identity. When a new observation enters with the same slotKey as an existing one, the existing observation is replaced in place (stable id, preserved anchor/createdAt/measurements). This guarantees at most one active observation per clinical slot in the live ledger.

Voice divergence risk: Voice observations get a level-suffixed conflictGroup (e.g. disc_contour_L4-L5) via ownershipFromObservation() in applyChangePlan.ts. The resolveConcept() function does NOT strip the suffix, producing a different slotKey than structured observations of the same clinical finding. This means a voice 'disc bulge at L4-L5' and a structured 'Disc bulge L4-L5' can coexist as two separate observations without same-slot replacement.

5. AI Report Composer Flow



Freshness / Staleness Mechanism (PR #656)

Two-axis validation: (1) reportRevision = hash of findingsHash + impressionHash + recommendationHash + obsCanon (clinically editable state). (2) inputHash = hash of studyCtxCanon + narrative + obsCanon + jobKindHint (full frozen AI input). When either differs from the stored enqueue-time value, READY flips to STALE_READY. inputHash is optional — legacy clients without it retain reportRevision-only behavior (backward compatible).

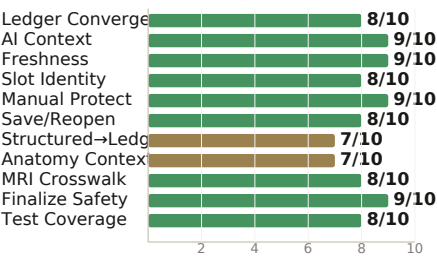
6. Safety & Finalization Gates

Gate	Location	What it Blocks	Status
Guard 3: Apply gate	useReportComposer.applyAccepted	STALE_READY + pending significant changes	SAFE - finalized
Guard 8: Frozen snapshot	computeSnapshotHashes + serverSnapshotHashes	Works as-is; never rereads live context	SAFE
Guard 9: Single Undo	store.undoLastPatch + lastPatchSnapshot	AS apply; not one undo step	SAFE
Guard 10: Finalize gate	workspace.tsx finalize flow	Finalized blocks all mutations (applyPathology)	SAFE - Overlay returns 'applied')
Manual text protection	patch.protected + provenance	Protected patches survive deselect	SAFE
Whole-format stale marking	confirmOverwriteAndApply	Prior patches marked stale (not deleted)	SAFE
AI Composer apply bypass	applyAiComposerAccepted	Writes narrative only; ledger NOT reconciled	RISK
Structured toggle-off	structuredObservations.ts	Does NOT remove ledger observation	GAP
Voice conflictGroup divergence	applyChangePlan.ts:ownership	ForceObservation conflictGroup produces different slotKey	RISK

7. Risk Matrix

Risk	Severity	Likelihood	Impact	Mitigation
AI Composer apply leaves ledger inconsistent	HIGH	MEDIUM	Re-compose loop; AI sees stale data	Add post-apply reconciliation: reconcilePatchAgainstLedger
Voice conflictGroup divergence (level mismatch)	MEDIUM	HIGH	Duplicate observations for same clinicalFindingKey	Remove {findingKey} suffix from ownershipFromObservation
Structured toggle-off doesn't remove observation	MEDIUM	HIGH	Ledger has orphan observations	Add structured toggle to removeObservation or manually remove
Persona router missing	MEDIUM	CERTAIN	Single generic prompt for all modalities/regions	modality/region persona/router.ts (documented as future change)
Quick Select tile ids non-deterministic	LOW	HIGH	Chip state not restored after reload	Use stable hash-based ids instead of uid()
root.exiting / root.traversing collapse	LOW	MEDIUM	Distinct nerve root semantics	Add separate concepts or document collapse
5712-line workspace monolith	LOW	CERTAIN	Hard to maintain; risky target for change	Extract into feature-scoped components

8. Maturity Chart



9. Test Coverage Summary

Test Suite	Files	Tests	Status
diagnostic-erp/src/lib/ (full tree)	214	2401	PASS
api-server reportComposer unit	1	13	PASS
api-server preDeploySafety	1	30	PASS
api-server pr656FreshnessHardening	1	20	PASS
api-server workerBootstrap	1	5	PASS
api-server request tests	1	(DB-gated skip)	ENV
Total	219	2469+	PASS

Key behavioral test files proving real mutation paths:

- structuredObservations.test.ts — 18 behavioral tests (A-T) for structured → ledger convergence
- composeObservations.test.ts — 33 tests for PR #654 AI adapter (canonical identity, dedupe, stale skip)
- composeStudyContext.test.ts — 21 tests for PR #656 context + inputHash freshness
- pr656FreshnessHardening.test.ts — 20 tests for dual-axis staleness (reportRevision + inputHash)
- sameSlotReplacement.test.ts — 14 tests for slot identity and same-slot replacement
- observationOwnership.matrix.test.ts — 8 tests for conflictGroup + concept resolution
- observationPersistence.hardening.test.ts — 8 tests for save/reopen ledger round-trip
- clinicalAnatomyContext.test.ts — 23 tests for anatomy catalog (clinic-format-driven)
- findingComposer.test.ts — 19 tests for Finding Composer model
- compositionWalkthrough.test.ts — 1 end-to-end walkthrough (5 golden reports)

10. Priority Roadmap

Priority	Task	Effort	Impact
P0	Add post-apply ledger reconciliation to applyAiComposeAccepted	S	Prevents re-compose loop + silent ledger divergence
P0	Unify voice conflictGroup (remove level suffix from ownershipFromObservation)	S	Structured observations same-slot replace correctly
P1	Implement persona router (persona/router.ts + persona/index.ts)	M	Per-modality/family system prompt specialization
P1	Add structured toggle-off → removeObservation or mark it stale	M	Ledger stays consistent with structured format state
P1	Add integration test for FormatApplyBridge round-trip	S	Format apply → region sync → studyContext propagation verified
P2	Extract RadiologyReportingWorkspace.tsx into feature-stopped component	M	Maintainability; safer targeted changes
P2	Cross-validate CONCEPT_CANON vs clinicalAnatomyContext concepts	S	Prevents concept drift between identity maps
P2	Make Quick Select tile ids stable (hash-based instead of uuid())	S	Chip selection state restored after reload
P2	Extend ABNORMALITY_HINTS to configurable list	S	New pathology types without code changes
P2	Add missing high-value Full Report Formats (Fazekas 2, Meningeal Enhancement, etc)	M	Complete clinical spec for common pathologies

11. What Should NOT Be Redesigned

- DO NOT REDESIGN:** ReportingStudyContext — the single canonical study-context resolver (reportingStudyContext.ts)
- DO NOT REDESIGN:** Observation Slot Identity — slotKey = region|concept|level|laterality (observationSlot.ts)
- DO NOT REDESIGN:** Canonical Observation Ledger — appliedPathologyPatches[] (store.ts)
- DO NOT REDESIGN:** applyPathologyOverlay — the single entrypoint for all observation mutations (store.ts:981)
- DO NOT REDESIGN:** PR #654 adapter — deriveComposeObservations feeds AI from the ledger (composeObservations.ts)
- DO NOT REDESIGN:** PR #656 freshness — inputHash + reportRevision dual-axis (snapshot.ts + useReportComposer.ts)
- DO NOT REDESIGN:** PR #657 persona — system prompt assembly (composeEngine.ts — to be EXTENDED, not replaced)
- DO NOT REDESIGN:** Full Report Format architecture — DEFAULT_REPORT_FORMATS + applyFormatById + mergeTwoFormats
- DO NOT REDESIGN:** Manual text protection — provenance-based; patch.protected flag; undoLastPatch
- DO NOT REDESIGN:** Finalize / Print / PDF / image pipeline — completely separate from observation concerns
- DO NOT REDESIGN:** Voice pipeline grammar — applyChangePlan.ts + voiceReportComposer/
- DO NOT REDESIGN:** AI jobs/Ollama runtime — resolveComposerRuntime + callOllama (no direct patient data)

12. Recommended Target Architecture

The current architecture is architecturally sound. The canonical observation ledger is the correct single source of truth for clinical observations. The PR #654/#656/#657 integration work (observations → AI Composer, canonical context → AI, persona rules → AI) is well-designed and should be preserved. The target architecture is the current architecture with the following targeted completions:

1. AI Apply Reconciliation: After applyAiComposerAccepted, run reconcilePatchAgainstNarrative for each patch. Mark stale when lastRendered.findings no longer appears in findingsText. Prevents re-compose loop.
2. Voice conflictGroup Unification: Remove `_${levelKey}` suffix from ownershipFromObservation. Let slotKey handle level via buildCanonicalObservation. Voice + structured same-slot replace correctly.
3. Structured Toggle-Off Removal: When a structured format toggle is turned off, call removeObservation for the corresponding structured-* observation. Or mark it stale.

4. Persona Router: Implement `persona/router.ts` + `persona/index.ts` (per-modality/family system prompt). Currently documented as future-work.
5. Component Extraction: Extract `RadiologyReportingWorkspace.tsx` (5712 lines) into `Composer/Voice/Format/Canvas/Coverage/Export` panels.

13. Final Verdict

NEEDS TARGETED HARDENING

Overall Maturity Score: 7.2 / 10

The CARE Radiology Reporting Workspace has a strong architectural foundation. The canonical observation ledger, slot identity, same-slot replacement, freshness protection, and AI Composer integration (PRs #654/#656/#657) are all well-designed and should be preserved. The structured reporting → ledger convergence (latest commit) is a significant milestone that closes the last major integration gap.

However, three targeted issues prevent a 'READY FOR CLINIC TEST' verdict:

P0-1: AI Composer apply bypasses the ledger. `applyAiComposerAccepted` writes narrative + provenance but does NOT create, update, or remove any `appliedPathologyPatch`. If the AI reorganizes observations, the ledger becomes inconsistent. Fix: add post-apply reconciliation via `reconcilePatchAgainstNarrative`.

P0-2: Voice conflictGroup divergence. Voice observations get a level-suffixed `conflictGroup` (`disc_contour_L4-L5`) while structured observations use plain concept (`disc_contour`). This produces different `slotKeys`, so the same clinical finding can coexist as two separate observations without same-slot replacement. Fix: remove the `_${levelKey}` suffix from `ownershipFromObservation`.

P0-3: Structured toggle-off leaves orphan observations. When a structured format toggle is turned off, the narrative text is correctly stripped, but the ledger observation remains active (not stale, not removed). The AI Composer still sees it. Fix: add structured-driven `removeObservation` or `mark-stale`.

Once these three P0 items are fixed, the workspace is ready for clinic test. The persona router (P1) and component extraction (P2) are quality improvements that do not block clinic safety.

Audit conducted on HEAD d4515dae (PR #658 branch, fix/mri-format-crosswalk-anatomy). No production code was modified during this audit. All findings are read-only observations derived directly from source files.